

Totally Integrated Automation Portal

Palitizer / PLC_1 [CPU 1211C DC/DC/DC] / Program blocks

FB_ModeManager [FB4]

FB_ModeManager Properties

General

Name	FB_ModeManager	Number	4	Type	FB	Language	SCL
Numbering	Automatic						

Information

Title		Author		Comment		Family	
Version	0.1	User-defined ID					

FB_ModeManager

Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
Input									
Output									
▼ InOut									
▼ io_Cmd	"UDT_SysCom-mand"			False	False	False	False		
xSwAuto	Bool			False	False	False	False		
xSwManual	Bool			False	False	False	False		
xBtnStart	Bool			False	False	False	False		
xBtnStop	Bool			False	False	False	False		
xBtnReset	Bool			False	False	False	False		
xEStop_OK	Bool			False	False	False	False		
xHmiResetCounters	Bool			False	False	False	False		
xHmiAckFault	Bool			False	False	False	False		
▼ io_Sts	"UDT_SysStatus"			False	False	False	False		
xModeAuto	Bool			False	False	False	False		
xModeManual	Bool			False	False	False	False		
xRunning	Bool			False	False	False	False		
xPaused	Bool			False	False	False	False		
xFaulted	Bool			False	False	False	False		
iStateID	Int			False	False	False	False		
iCurrentLayer	Int			False	False	False	False		
iTargetLayers	Int			False	False	False	False		
iTotalBoxes	DInt			False	False	False	False		
iTotalPallets	DInt			False	False	False	False		
wAlarmWord	Word			False	False	False	False		
Static									
Temp									
Constant									

```
0001 // --- 1. Master Safety Override ---
0002 // If the E-stop is pressed, immediately kill the run and trigger a fault.
0003 IF NOT #io_Cmd.xEStop_OK THEN
0004     #io_Sts.xFaulted := TRUE;
0005     #io_Sts.xRunning := FALSE;
0006 END_IF;
0007
0008 // Fault reset logic
0009 IF #io_Cmd.xBtnReset AND #io_Cmd.xEStop_OK THEN
0010     #io_Sts.xFaulted := FALSE;
0011 END_IF;
0012
0013 // --- 2.Mode selection ---
0014 // Determine machine mode based on physical switch
0015 IF #io_Cmd.xSwAuto AND NOT #io_Sts.xFaulted THEN
0016     #io_Sts.xModeAuto := TRUE;
0017     #io_Sts.xModeManual := FALSE;
0018
0019 ELSIF #io_Cmd.xSwManual AND NOT #io_Sts.xFaulted THEN
0020     #io_Sts.xModeAuto := FALSE;
0021     #io_Sts.xModeManual := TRUE;
0022     #io_Sts.xRunning := FALSE; // Instantly drop the run permissive if dropping out of Auto
0023
0024 ELSE
0025     #io_Sts.xModeAuto := FALSE;
0026     #io_Sts.xModeManual := FALSE;
0027     #io_Sts.xRunning := FALSE;
0028 END_IF;
0029
0030 // --- 3. Start / Stop Latch (Auto Mode Only) ---
0031 IF #io_Sts.xModeAuto THEN
0032     // start sequence
0033     IF #io_Cmd.xBtnStart THEN
0034         #io_Sts.xRunning := TRUE;
0035     END_IF;
```

Totally Integrated Automation Portal

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0036
0037 // Hold sequence pause
0038 IF NOT #io_Cmd.xBtnStop THEN
0039   #io_Sts.xRunning := FALSE;
0040 END_IF;
0041
0042 END_IF;
0043
0044 // --- 4. HMI State & Indicator Logic ---
0045 IF #io_Sts.xFaulted THEN
0046   #io_Sts.iStateID := 5; // Faulted
0047 ELSIF #io_Sts.xModeManual THEN
0048   #io_Sts.iStateID := 4; // manual mode
0049 ELSIF #io_Sts.xModeAuto AND #io_Sts.xRunning THEN
0050   #io_Sts.iStateID := 2; // auto mode running
0051 ELSIF #io_Sts.xModeAuto AND NOT #io_Sts.xRunning THEN
0052   #io_Sts.iStateID := 3; // auto mode paused
0053 ELSE
0054   #io_Sts.iStateID := 1; // idle
0055 END_IF;
0056
0057 // Indicator Light Flag (TRUE when paused in Auto or waiting in Manual)
0058 #io_Sts.xPaused := (#io_Sts.iStateID = 3) OR (#io_Sts.iStateID = 4);
```

Symbol	Address	Type	Comment
#io_Cmd.xBtnReset		Bool	
#io_Cmd.xBtnStart		Bool	
#io_Cmd.xBtnStop		Bool	
#io_Cmd.xEStop_OK		Bool	
#io_Cmd.xSwAuto		Bool	
#io_Cmd.xSwManual		Bool	
#io_Sts.iStateID		Int	
#io_Sts.xFaulted		Bool	
#io_Sts.xModeAuto		Bool	
#io_Sts.xModeManual		Bool	
#io_Sts.xPaused		Bool	
#io_Sts.xRunning		Bool	